Daily Research Digest - 2026-04-07

Topics: galaxy clusters, galaxies, gravitational lensing, dark matter

Paper Summaries and Figures

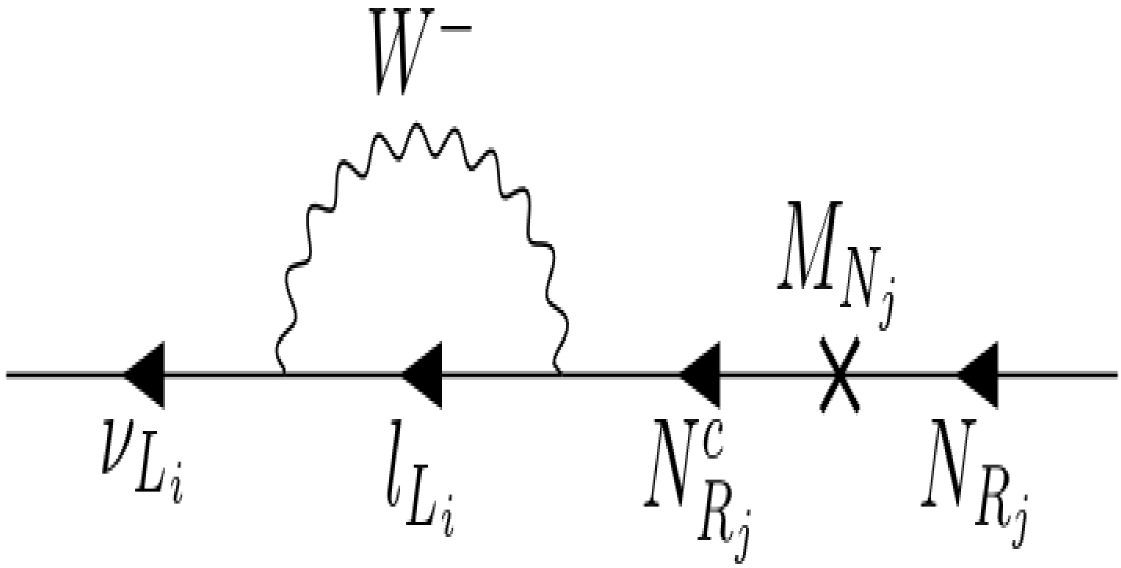
1. Light neutrinos, Dark matter and leptogenesis near electroweak scale and Z_4 symmetry

Focus: dark matter
Authors: Kunal Pandey, Rathin Adhikari
Published: 2026-04-06T17:33:46+00:00
Summary: Considering Z_4 symmetry in Type I seesaw scenario, one could obtain mass-squared differences of light neutrinos, mixings and CP violating phase within 3 σ confidence level based on neutrino oscillation data. This is possible with only three independent complex parameters for allowed Yukawa couplings and one real mass parameter for heavy right handed neutrino fields around electroweak scale. After considering only three more real parameters as coming from small soft-symmetry breaking terms, the lightest right handed neutrino could be considered as dark matter candidate via freeze-in mechanism and the other two heavier right handed neutrinos through their decays, could generate the baryonic asymmetry of the universe naturally via resonant leptogenesis.

Page-by-page summaries

Page 1: The paper explores how Z4 symmetry in the Type I seesaw scenario can explain light neutrino mass-squared differences, mixings, and CP-violating phase with just three complex Yukawa couplings and one real heavy right-handed neutrino mass parameter near the electroweak scale. Additionally, introducing three small soft-breaking terms allows the lightest right-handed neutrino to serve as dark matter via freeze-in mechanism, while resonant leptogenesis from heavier right-handed neutrinos decay generates baryonic asymmetry.
Page 2: The page introduces the challenge of accommodating light neutrino masses within the Standard Model through the Type-I seesaw mechanism, which typically requires heavy right-handed neutrinos at a scale of ~109 GeV. It discusses various approaches to lowering this scale, including massless textures with loop corrections and additional scalar interactions. The paper focuses on textures that yield three massless neutrinos at tree level without fine-tuning, motivated by a preserved Z4 symmetry in field interactions.

Selected figures



2. Measurement of the galaxy-velocity power spectrum of DESI tracers with the kinematic Sunyaev-Zeldovich effect using DESI DR2 and ACT DR6

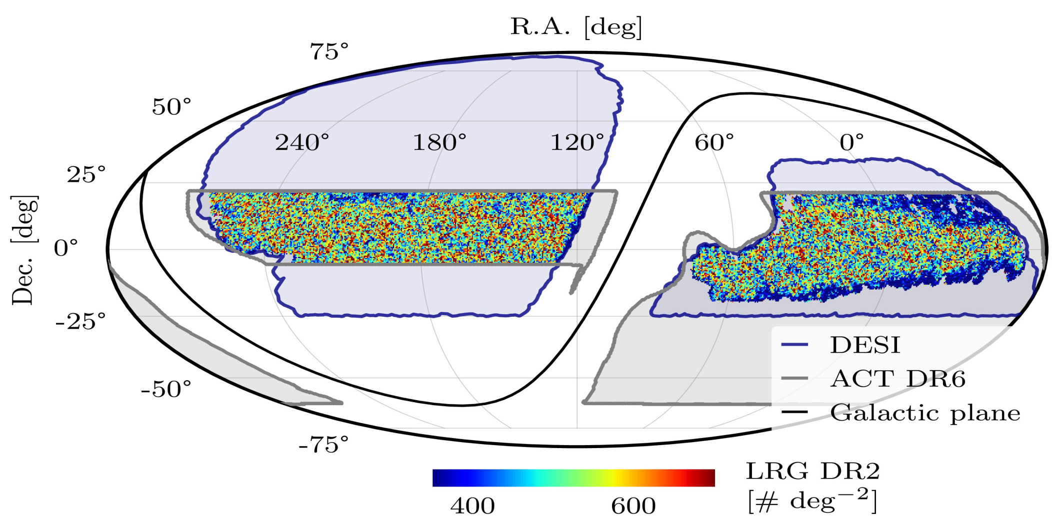
Focus: galaxies
Authors: Edmond Chaussidon, Selim C. Hotinli, Simone Ferraro, Kendrick Smith, Xinyi Chen, J. Aguilar, S. Ahlen, D. Bianchi, D. Brooks, T. Claybaugh, A. Cuceu, A. de la Macorra, B. Dey, P. Doel, A. Font-Ribera, J. E. Forero-Romero, E. Gaztañaga, S. Gontcho A Gontcho, G. Gutierrez, J. Guy, H. K. Herrera-Alcantar, K. Honscheid, C. Howlett, D. Huterer, M. Ishak, R. Joyce, D. Kirkby, A. Kremin, O. Lahav, M. Landriau, L. Le Guillou, M. Manera, A. Meisner, R. Miquel, S. Nadathur, J. A. Newman, N. Palanque-Delabrouille, W. J. Percival, F. Prada, I. Pérez-Ràfols, G. Rossi, L. Samushia, E. Sanchez, D. Schlegel, M. Schubnell, H. Seo, J. Silber, D. Sprayberry, G. Tarlé, B. A. Weaver, C. Yèche, R. Zhou
Published: 2026-04-06T17:16:00+00:00
Summary: Joint analyses of high-resolution CMB temperature maps with galaxy surveys provide a unique way to reconstruct the radial velocity field of the underlying matter distribution via the kinematic Sunyaev-Zeldovich (kSZ) effect. Using data from the Atacama Cosmology Telescope (ACT) DR6 and the Dark Energy Spectroscopic Instrument (DESI) DR2, we present radial velocity reconstructions for luminous red galaxies (LRGs), emission-line galaxies (ELGs), and quasars (QSOs). Leveraging the spectroscopic data, we are able to reliably model the foreground contamination and report a negligible impact on our main observables. We detect the velocity-galaxy cross-correlation at 17.0σ for LRGs, and for the first time, at 8.3σ for ELGs and 6.8σ for QSOs. We further report the first detection of the velocity-velocity correlation using LRGs at 3.1σ, as well as the highest cumulative detection of the kSZ effect to date at 20.8 σ. Similarly to previous results, we find a lower amplitude of the kSZ signal compared to our fiducial halo model prediction and electron profile assuming a Battaglia profile. Combining these new observables, we obtain constraints on local-type primordial non-Gaussianity (PNG): f_ NL^ loc = 15.9_-34.4^+34.6 at 68\% confidence, which represents the tightest constraint to date derived from the velocity field. The measurements presented here already exhibit lower noise on a per-mode basis than the galaxy auto-correlation on the largest scales, k<0.004~Mpc^-1, highlighting the key role these observables will play in the context of future CMB experiments such as the Simons

Observatory.

Page-by-page summaries

Page 1: The research paper presents a collaborative effort to measure the galaxy-velocity power spectrum using DESI DR2 and ACT DR6 data, leveraging the kinematic Sunyaev-Zeldovich effect. The authors, hailing from multiple institutions worldwide, aim to provide insights into cosmic structures and their dynamics through this comprehensive analysis.
Page 2: The page discusses the use of joint analyses of high-resolution CMB temperature maps from ACT DR6 with DESI DR2 data to reconstruct radial velocity fields for LRGs, ELGs, and QSOs via the kinematic Sunyaev-Zeldovich effect. The study reports significant detections of velocity-galaxy cross-correlation for all three galaxy types and the first detection of velocity-velocity correlation using LRGs, achieving the highest cumulative kSZ effect detection to date at 20.8σ. These measurements also provide competitive constraints on local-type primordial non-Gaussianity (PNG) with f loc NL = 15.9+34.6 –34.4 at 68% confidence.

Selected figures



3. Probing cosmic anisotropy with galaxy clusters and supernovae

Focus: galaxy clusters
Authors: Shubham Barua, Sujit K. Dalui, Shantanu Desai
Published: 2026-04-06T04:19:03+00:00
Summary: Using ΛCDM and Padé-(2,1) cosmography, we study directional variations in the Hubble constant, H_0, using galaxy cluster and Type Ia Supernovae (from Pantheon Plus) by the hemisphere decomposition method. Since there is a degeneracy between H_0 and absolute magnitude M_B for Supernovae, Cepheid host calibration is usually required to constrain H_0. Hence, in this work in order to complement the Cepheid host calibration in Supernovae, we also use calibrations based on galaxy cluster scaling relations. We find that there is a 1σ difference in H_0 variations when using galaxy

clusters as calibrators compared to Cepheids highlighting that the variations in H_0 are robust across different calibration methods. Across all combinations of models and data sets used, we obtain a consistent deviation 2σ from isotropy. In nearly all cases, we notice that the maximum ΔH_0 aligns with the CMB dipole direction.

Page-by-page summaries

Page 1: The authors investigate directional variations in the Hubble constant H_0 using galaxy clusters and Type Ia Supernovae data sets, applying ΛCDM and Padé-(2,1) cosmography models. They find a consistent ~2σ deviation from isotropy across all combinations of models and data sets, with maximum H_0 aligning with the CMB dipole direction, suggesting potential anisotropies in the universe that could affect the measurement of H_0.
Page 2: Page 2 outlines the distribution and properties of galaxy clusters (GCs) and Type Ia supernovae (SNe), showing that GC data from Chandra and XMM-Newton are relatively uniform across the sky, while SNe exhibit inhomogeneous distribution with a belt-like structure at high redshifts. The page also details the LX–T scaling relation for GCs and provides χ2 expressions to quantify uncertainties for both GCs and SNe datasets.

4. Raychaudhuri Equation and Weyl-Driven Shear: A Weak-Field Approach to Lensing and Gravitational Waves

Focus: gravitational lensing
Authors: Madhukrishna Chakraborty, Subenoy Chakraborty
Published: 2026-04-05T06:32:45+00:00
Summary: The letter studies phenomena like gravitational wave propagation and gravitational lensing using the celebrated Raychaudhuri equation (RE) in the weak field limit. Newtonian analogue of Relativistic RE has been explored. In doing so, role of shear has been found to be extremely important in explaining these phenomena. Consequently, the RE for shear has been used in course of the study and importance of Weyl curvature tensor in lensing and gravity wave propagation has been explicitly shown using a damped harmonic oscillator approach.

Page-by-page summaries

Page 1: The letter explores how gravitational waves and weak gravitational lensing can be described using the Raychaudhuri Equation (RE) in the weak-field limit, focusing on the role of shear governed by the Weyl curvature tensor. It introduces a damped harmonic oscillator approach to unify the evolution of shear in both phenomena, providing a covariant and gauge-invariant description.
Page 2: The page introduces the Raychaudhuri Equation (RE) in general relativity, detailing its role in describing the evolution of geodesic congruences through spacetime. It breaks down the kinematic variables—expansion scalar θ, shear tensor σab, vorticity tensor ωab, and 4-acceleration Aa—and presents the REs for both time-like and null geodesics, emphasizing how these equations govern the focusing, shearing, and twisting of geodesic bundles.